

Architectural Pattern Selection & Justification Document

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Domain & Problem Statement: Healthcare & Telemedicine — #19 Personalized Meal & Diet Subscription Manager

Architecture Selection

“We chose Microservices Architecture for the Personalized Meal & Diet Subscription Manager System.”

1. Architectural Style Comparison Matrix

Architectural Style	Structure	Fault Tolerance	Scalability	Suitability for Problem #19
Layered Architecture	Rigid horizontal tiers	Low (Monolithic failure)	Monolithic scaling	Poor (High coupling between solver & UI)
Microservices Architecture (Selected)	Autonomous services	High (Fault isolation)	Independent auto-scaling	Optimal (Decouples PHI, batch solver & delivery)
Client-Server Architecture	Centralized server	Single point of failure	Bottleneck under peak load	Poor (Central DB exposes PHI records)

2. Two Scenario-Related Reasons for Architectural Choice

2.1 Independent Elastic Scaling (Batch Meal Solver vs. Real-Time UI)

The **Dietary Engine Component** executes computationally intensive combinatorial optimization algorithms to generate weekly menus adhering strictly to macro targets and allergen exclusion lists (FR-001) on periodic batch cycles.

Conversely, the **Subscription & Order Manager** handles continuous real-time traffic.

Microservices allow independent auto-scaling for the meal solver without over-provisioning lightweight web API services.

2.2 Fault Isolation Across Third-Party Integrations

Integrations with external **Payment Gateway API** and **Delivery System API** introduce potential third-party downtime.

Circuit breaker patterns isolate gateway failures—ensuring Subscribers can still modify health profiles and Dietitians can approve menus even during payment or delivery gateway outages.

3. Security Advantage

Encapsulation and PHI Isolation (HIPAA/GDPR Compliance)

Patient health profiles (diabetic limits, allergen lists) represent sensitive PHI.

Encapsulating this inside an isolated **User & Health Profile Manager Component** restricts direct database access.

External services (Payment, Delivery) communicate solely via OAuth2/JWT authenticated REST/gRPC endpoints with Role-Based Access Control (RBAC), preventing third-party access to clinical health records.

4. Performance Benefit

In-Memory Caching & Event Streaming for Schedule Modifications (NFR-001 Support)

Supporting schedule modifications up to 12 hours before dispatch under peak traffic (NFR-001) is achieved by routing requests to asynchronous event queues (Kafka/RabbitMQ) and localized Redis caches inside **Subscription & Order Manager**.

This bypasses lock-heavy database disk transactions, guaranteeing **<150ms response times**.